Laplace Transform

Based on Chap 9 of OWN.

1. Quiz 6 : 20 April

2. Exam 2 : 30 April 6:00 pm

I- Introduction

Given a signal $x(t)$, we would like to find

$$X(s) \triangleq \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} \, dt \, , \quad s \in \mathbb{C}$$

$$e^{st} \longrightarrow \boxed{h} \longrightarrow e^{st} \cdot H(s).$$

$$H(s) = \int_{-\infty}^{+\infty} h(t) \cdot e^{-st} \, dt.$$

In general, $X(s)$ is well-defined for only certain values of s. Suppose $s = \sigma + j\omega$, $\sigma, \omega \in \mathbb{R}$, we integrate

$$x(t) \cdot e^{-st} = x(t) \cdot e^{-(\sigma + j\omega)t}$$

$$= x(t) \cdot e^{-\sigma t} \cdot e^{-j\omega t}.$$

If $\int_{-\infty}^{+\infty} |x(t) e^{-st}| \, dt$ is finite, then:

$$|X(s)| = \left| \int_{-\infty}^{+\infty} x(t) \cdot e^{-st} \, dt \right| \leq \int_{-\infty}^{+\infty} |x(t) \cdot e^{-st}| \, dt, \text{ and}$$

hence $|X(s)| < \infty$ and hence $X(s)$ is well-defined.

We will restrict ourselves to such choices of $s \in \mathbb{C}$ so that

$x(t) \cdot e^{-st}$ is absolutely integrable.

The Region of Convergence (ROC) of $x(t)$ is defined to be the set of values of s such that

$$\int_{-\infty}^{+\infty} \left| x(t) \cdot e^{-st} \right| dt \text{ is finite.}$$

If $s = \sigma + j\omega$, $\quad \left| x(t) \cdot e^{-st} \right| = \left| x(t) \cdot e^{-\sigma t} \cdot e^{-j\omega t} \right|$

$$= \left| x(t) \cdot e^{-\sigma t} \right|.$$

$$\therefore \int_{-\infty}^{+\infty} \left| x(t) \cdot e^{-st} \right| dt = \int_{-\infty}^{+\infty} \left| x(t) \cdot e^{-\sigma t} \right| dt.$$

Hence, whether $s \in$ ROC depends only on its real part, and not on the imaginary part.

Notation:

$$x(t) \xleftrightarrow{L} \underbrace{X(s), \quad \text{ROC}}_{\text{You MUST specify both}}.$$

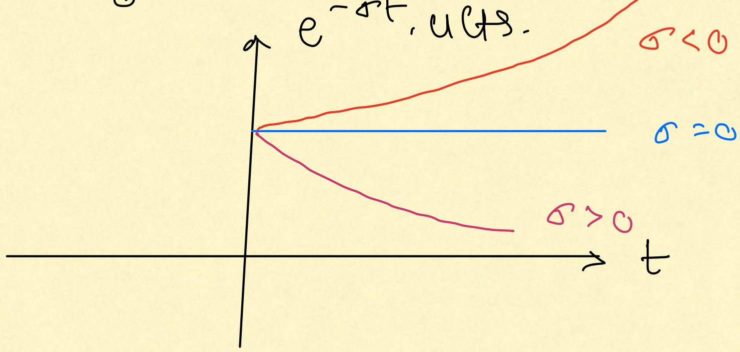
II. Examples.

(1) $x(t) = u(t)$.

(a) ROC:

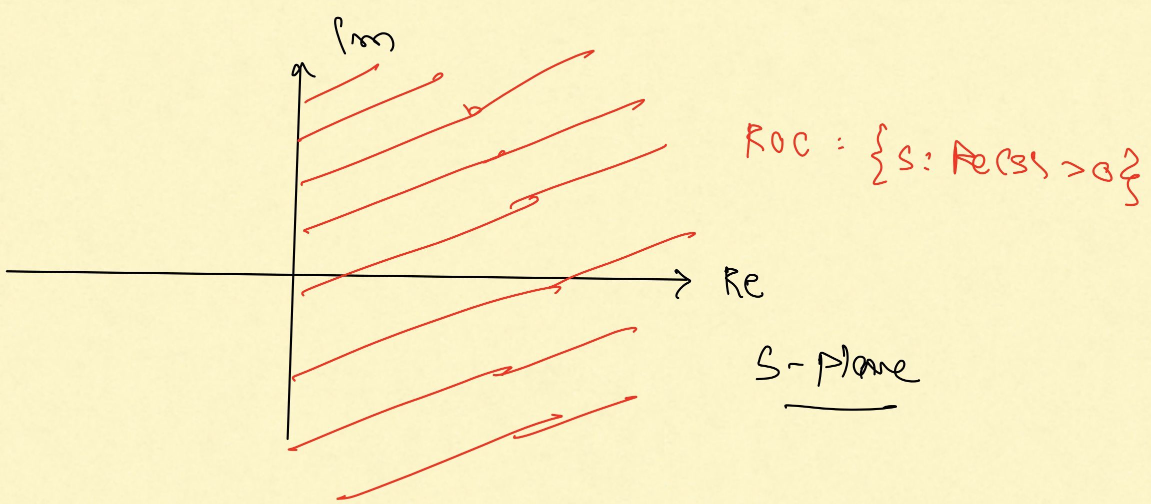
$s \in$ ROC if $\displaystyle\int_{-\infty}^{+\infty} |x(t)\cdot e^{-\sigma t}|\,dt$ is finite.

$$\int_0^{\infty} |u(t)\cdot e^{-\sigma t}|\,dt = \int_0^{\infty} e^{-\sigma t}\,dt.$$



$\therefore$ $s \in$ ROC iff $\sigma > 0$ i.e. $Re(s) > 0$.

$\therefore$ ROC $= \{ s : Re(s) > 0 \}$.

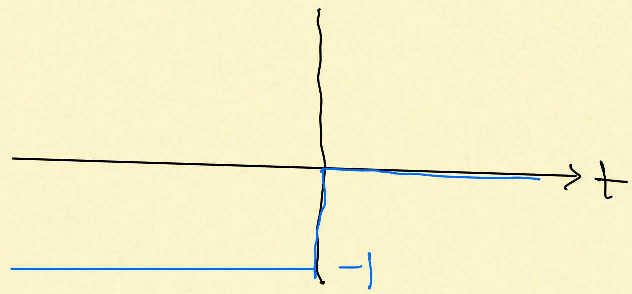


(b) Assume $s \in$ ROC i.e. $Re(s) > 0$.

$$X(s) = \int_0^{\infty} e^{-st}\,dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} = \frac{1}{s}.$$

$$u(t) \xleftrightarrow{L} \frac{1}{s}, \quad Re(s) > 0.$$

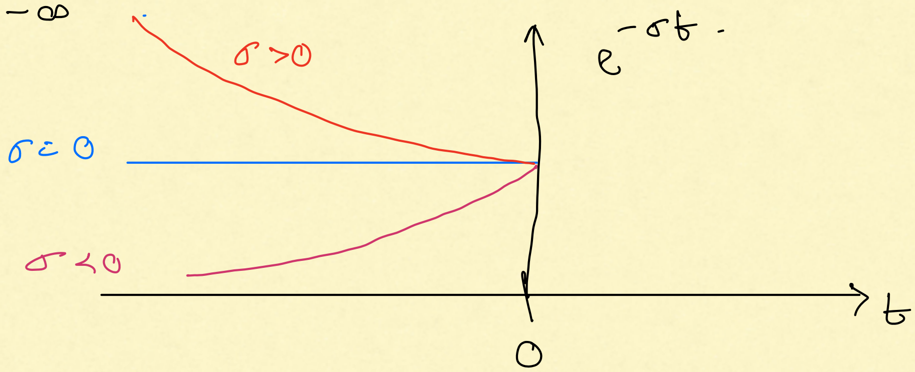
(2) $x(t) = -u(-t)$



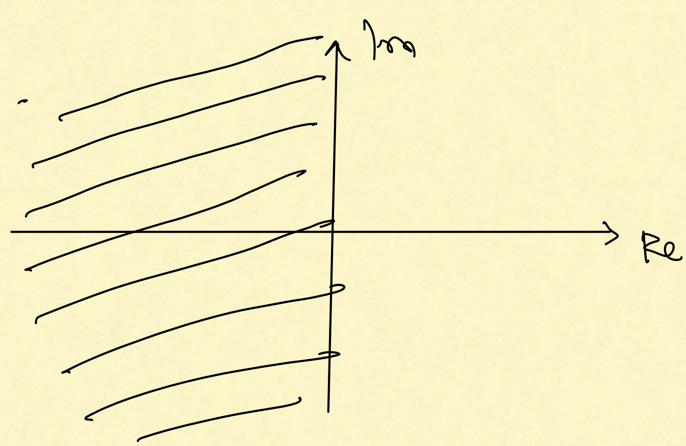
(a) $s \in ROC$ iff

$$\int_{-\infty}^{+\infty} |e^{-st} \times -u(-t)| \, dt \text{ is finite.}$$

i.e. $$\int_{-\infty}^{0} e^{-\sigma t} \, dt. \text{ is finite.}$$



$ROC = \{ s : Re(s) < 0 \}$.



(b) Assume $s \in ROC$:

$$X(s) = \int_{-\infty}^{0} (-1) \times e^{-st} \, dt = \frac{1}{s}$$

$\therefore \quad -u(-t) \xleftrightarrow{L} \frac{1}{s}, \quad Re(s) < 0$

Compare with $\quad u(t) \xleftrightarrow{L} \frac{1}{s}, \quad Re(s) > 0$

<u>Remark:</u> Suppose $y(t) \xleftrightarrow{F} Y(\omega)$

Under certain conditions it is possible to retrieve $y(t)$ from $Y(\omega)$.

Consider $y(t) = x(t) \cdot e^{-\sigma t}$.

$y(t) \cdot e^{-j\omega t} = x(t) \cdot e^{-\sigma t} \cdot e^{-j\omega t} = x(t) \cdot e^{-st}$.

Then:
$$X(s) = \int_{-\infty}^{+\infty} x(t) \cdot e^{-\sigma t} \cdot e^{-j\omega t}\, dt$$

$$= \int_{-\infty}^{+\infty} y(t) \cdot e^{-j\omega t}\, dt.$$

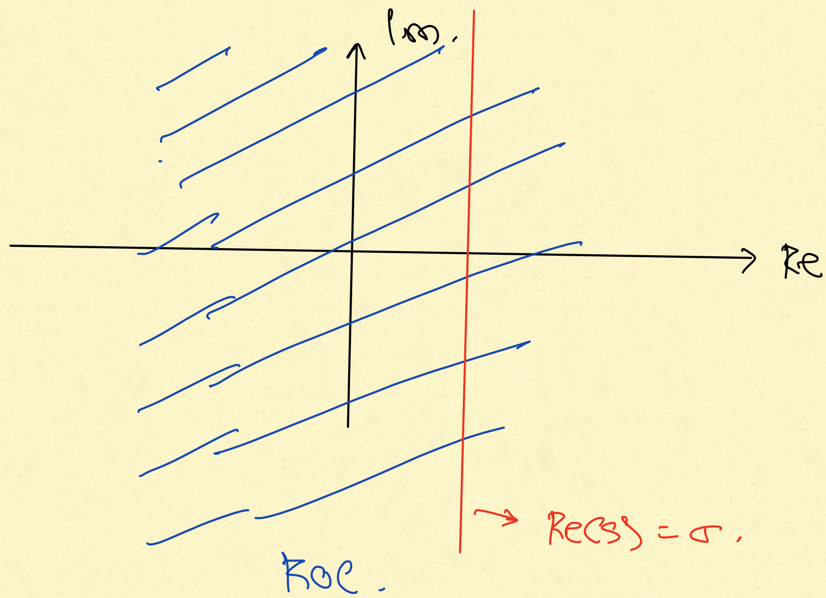
$$= Y(\omega).$$

$\therefore$ If $y(t)$ satisfies the conditions for inverting the fourier transform:

$$y(t) = \frac{1}{2\pi} \int_{\omega=-\infty}^{+\infty} Y(\omega) \cdot e^{j\omega t}\, d\omega$$

$$x(t) \cdot e^{-\sigma t} = \int_{-\infty}^{+\infty} \frac{Y(\omega) \cdot e^{j\omega t}}{2\pi}\, d\omega.$$

$$\therefore \quad x(t)\cdot e^{-\sigma t} = \frac{1}{2\pi}\int_{\omega=-\infty}^{+\infty} X(\sigma + j\omega)\cdot e^{j\omega t}\, d\omega$$



$$\therefore \quad x(t) = \frac{e^{\sigma t}}{2\pi}\int_{\omega=-\infty}^{+\infty} X(\sigma + j\omega)\cdot e^{j\omega t}\, d\omega.$$